



Uber's Bug Bounty Prog

Uber announced its bug bounty program and is using HackerOne as the platform.
<http://dgit.in/UberBBP>



Throttle goes free

Throttle, a browser plugin that keeps spam away from your inbox goes completely free.
<http://dgit.in/ThrottlePlgn>



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Where innovation meets tech

Do you love gadgets and gizmos? Ever dreamt of creating your own gadgets? Or maybe you're just an electronics enthusiast who likes to rip everything apart and learn how things work inside circuits. Well, look no further because here's a way you can really hack into stuff and experiment and build cool devices.



Sonali KM

So now before we start, we'll just give an introduction to what we're going to talk about. Ever heard of 'Open Source Software'? If you haven't, it's just software that you get the full source code to so that you can tweak it the way you want and it's for free and there's no paid license to it so you can really play around with it to suit your needs. The only condition is that you need

to give away whatever changes you've made to the source code. An example is the Linux operating system.

Now there's something called 'open source hardware'. It's similar to open source software in that you can alter it the way you want and build something cool and the circuit diagrams like the source code are available to you. We'll give an example of PhoneBlocs. It's a smartphone that you can 'edit' the way you want. If you want a better camera for your phone, instead of purchasing a new phone with a much more superior camera quality, you can just purchase a block that contains the camera and fit it into your pre-existing phone using PhoneBlocs.

How do I build my own stuff?

By using Open Source Computing Hardware!

There's this little kid called the Arduino. Well, get him home first. You can either purchase an Arduino starter kit or buy the parts separately. We bought a set with an Arduino Uno, a keypad and an LCD display with a USB cable to connect the Uno to our laptop, from Amazon, and it worked out for quite cheap. We had other electronic stuff that we had already purchased when this hobby idea came up.

But how do you build stuff with it? Before we say that, we'll explain to you what an Arduino board is. It's just a small circuit board with a built-in microcontroller (MC) by Atmel that's either 8, 16 or 32-bit (currently all models are the latter version) and contains memory and digital pins to control the MC chip. There's a USB cable to connect the board to your computer. You have to download an IDE (Integrated Development Environment) to be able to code and upload the program you have

written into your board. For this, Atmel is providing its own IDE but the preferred and most used one is the Arduino IDE that's freely downloadable.

Now purchase or just download the eBook required for building stuff with your Arduino. We suggest '*Beginning Arduino*' by Michael McRoberts and '*Programming Arduino-Getting Started with Sketches*' by Simon Monk. There's also this cool book called '*Getting Started with Arduino*' by Massimo Banzhi who is the co-founder of Arduino. He really teaches you how 'everything electronics' works and reignites the tech enthusiast in you who might have one day or the other dissected an old gadget at home.

What can I build?

To start off with, you'll learn how to blink an LED first that is built into the board. It's like a 'Hello World' program that you write when you're starting out on learning a new programming language. You can also try attaching your own LED to the board and make that blink. Then you move onto creating an SOS signal. Then traffic lights. Then making a keypad work. An LCD screen displaying your messages. An alarm system. And lots more!

As you progress, you'll learn how to build a water level detector, a temperature sensor and the project that we would love to build soon, your very own pet robot!

We also managed to build a distance sensor that displays warning LEDs and produces various sounds using a buzzer to indicate how far you are from an obstacle when you're reversing your car. Did I mention you can control the lights, ventilator fans, irrigation system and even sun shades by just using the Bluetooth feature of your phone? Just buy a small Bluetooth module and connect it to your circuit and it will do all the work for you controlled by an app on your smartphone.

Open source computing hardware...Where's the computer?

This is where we'll introduce you to the Raspberry Pi. It's similar to the Arduino except that it's not just a MC and memory. It's an entire computer CPU! But how can it be when it's also just a

